An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Swastik Kumar · Electronics & Communication Engineering
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1. Executive summary

Floods displace millions of people in India each year, and the limiting factor in a rescue is not reaching survivors but locating them.

We propose three one-metre, 6.4 kg multirotor aircraft that search flooded ground as a coordinated group under a single operator, identify people from the air, report their position to approximately one metre, and deliver a relief kit to each. The system requires no mobile network and no internet connection.

The sizing, error budgets and flight software are complete and reproducible, and coordination has been demonstrated across three simulated autopilots. No aircraft has yet flown, so the figures presented here are calculated; the programme below is ordered to replace them with measurements, beginning with the most consequential.

The request is **INR 7,43,004 across six phases and 32 steps**, no step exceeding INR 30,000, each released on the demonstrated result of the preceding one. The first is INR 31,371.

2. The problem

After a flood, the binding constraint is time-to-locate. Boat teams search slowly, at considerable risk, and cannot observe rooftops. Helicopters are scarce, costly per flight hour, and generally committed to evacuation. Neither covers a large area quickly while reporting precise positions.

Conventional small drones do not close this gap. They require one pilot per aircraft, depend on network infrastructure the disaster has removed, and locate survivors without assisting them.

The more difficult problem is detection itself. A person in floodwater presents only head and shoulders, approximately 40 cm across. Detection models accept a fixed, small input, so the conventional approach reduces the image to fit — and the target reduces with it, to a size at which published results report accuracy between 29 % and 61 %. Selecting a smaller, faster model worsens this, because smaller models accept smaller inputs. Two further constraints are specific to flooding: no existing dataset annotates *people* in flood imagery, and thermal imaging is ineffective in water, where skin temperature equilibrates toward that of the water.

3. Our solution

System architecture. The operator defines a search region of arbitrary shape; the software partitions it into three balanced sub-regions and plans a coverage path over each. This has been verified across thirteen non-convex boundary shapes with no path leaving the region and workload imbalance below 0.02 %. The aircraft then operate autonomously, and the test harness demonstrates structurally that no command is transmitted after the mission begins. Detection and geolocation execute on board, so neither depends on a radio link. Each aircraft carries four kits and can descend, null its groundspeed over a survivor, and release one. Link loss, low battery and operator abort all converge on a single return-and-land behaviour.

Detection approach. The image is not reduced. Each full-resolution frame is divided into 48 overlapping tiles processed at native resolution, and the survey altitude is 40 m rather than 60 m. Together these place the target at **39 pixels** rather than 13 — nine times the area, and outside the size band in which detector performance degrades sharply. Both changes are required; either alone leaves the target within that band.

The approach requires four times the computation, and this has been verified as feasible before adoption: 48 tiles at 2 Hz is 96 inferences per second, against 130–160 measured for the selected accelerator. The additional transects at the lower altitude cost seconds, and total mission duration is marginally shorter than at 60 m because the climb is reduced.

One item remains open. Confirming a survivor across twelve frames requires approximately 3 Hz, which exceeds the accelerator's capacity. A two-stage approach would recover this, using a low-cost model to eliminate the approximately 87 % of tiles containing only water. It will be evaluated against public data before adoption, since a first-pass filter that discards a tile containing a person fails without indication.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that displays mission state and is architecturally incapable of commanding the aircraft, so autonomy cannot be circumvented in use.
3. Autonomy software — area partitioning, path planning, task allocation and failsafe logic — with an automated test suite.
4. A measured detection accuracy figure, replacing the calculated one.
5. A measured determination of whether the two-stage filter permits the capture rate required for twelve-frame confirmation.
6. A publication: image tiling, aerial person detection and low-power onboard inference have not previously been combined for flood conditions.
7. Source code and results published in either outcome.

5. Benefits

Operational. Three aircraft survey ten hectares in under eight minutes and provide a rescue coordinator with a list of coordinates rather than a video feed requiring continuous attention, and do so without network infrastructure.

Economic. The complete system costs less than one hour of helicopter search time, is transportable by road, and is operated by two people.

Institutional. A calibrated thrust stand, a certified scale, a ground station, a tested autonomy codebase and trained students remain with the department after the project concludes.

Strategic. Indian suppliers are selected wherever they meet the specification, both for supply-chain sovereignty and because domestic lead times are approximately half those of imports, which directly increases available flight-test time.

Transferable. Only the training data is flood-specific. The same system applies to earthquake rubble survey, wildfire perimeter mapping and avalanche search.

6. Resources needed

The programme is divided into six phases. Each has a stated objective, a defined completion criterion, and is released only when the preceding phase has demonstrated its result. No individual step exceeds INR 30,000.

Every component below is a specific model at a current Indian retail price, selected against a quantity the design produces. Where a lower-cost alternative was considered and rejected, the reason is stated.

Phase 1 — Propulsion verification steps 1–2 INR 31,371

Objective: determine whether the selected motor produces the thrust the design requires, and establish a mass reference against which the build can be tracked. **Released:** on approval. **Completion criterion:** measured static thrust of $\geq$3.18 kgf per motor at full throttle.

Item	Model / specification	INR	Selection rationale
Calibrated scale	Bench scale, 30 kg capacity, 1 g resolution, with calibration certificate	12,000	Maximum take-off mass is a hard limit. A resolution of 1 g against a 6 360 g aircraft is 0.02 %, permitting mass to be tracked component by component during assembly. Certification is required for the resulting figure to be defensible.
Thrust stand	20 kg load cell, HX711 24-bit amplifier, inline power meter, frame fabricated in-house	8,000	The selected motors are supplied without a published thrust curve, so the governing quantity of the design is presently unverified. Simultaneous thrust and current measurement yields thrust per watt, which determines endurance. Commercial instruments begin near INR 45,000; the sensing components cost under INR 1,000 and the frame is within the institute machine shop's capability.
Motor (single)	Tarot TL96020, 5008 frame, 340 KV	4,500	One unit is procured for measurement before the remaining eleven are committed. The requirement is 3.18 kgf at full throttle with hover at approximately half, where propeller efficiency is highest and control authority remains available for gust rejection.

If the measurement fails, the programme concludes at INR 31,371, and both instruments remain departmental assets.

Phase 2 — First aircraft steps 3–10 INR 1,97,238

Objective: construct and integrate one complete aircraft capable of autonomous flight, onboard detection and payload release. **Released:** on the Phase 1 thrust measurement. **Completion criterion:** airframe assembled, weighed, and all avionics powered and communicating.

Item	Model	INR	Selection rationale
Flight controller	Holybro Pixhawk 6C Mini	22,600	Two independent inertial measurement units on silicone isolation, so a single sensor failure or airframe resonance does not terminate the flight. Runs ArduPilot natively, which the simulation testing is built against. F4-class boards at approximately INR 6,000 lack the memory required for the logging and failsafe configuration in use.
GNSS receiver	RTK-capable, RTCM3 corrections, $\leq$3 cm CEP	18,000	Positioning accuracy is the largest single error term in the system. Standard GNSS yields a reported survivor position accurate to approximately 3.9 m; RTK yields 0.75 m. No other expenditure improves this figure comparably.
Onboard computer	Raspberry Pi 5, 8 GB	8,000	Hosts the operating system, coverage planner, message routing, mesh link and delivery logic. It is also the only board in this class exposing PCIe, which the accelerator requires; the preceding generation is not compatible.
AI accelerator	Raspberry Pi AI HAT+ 26 TOPS (Hailo-8)	11,000	Executes the detector on board, so detection is independent of the radio link. Measured at 130–160 frames per second for the model and input size in use, against a requirement of 96. The 13 TOPS variant is approximately INR 4,000 lower but achieves 60–80, which is insufficient.
Camera and lens	Arducam IMX477 with 6 mm CS lens, fixed focus	11,100	A 1.55 µm pixel pitch yields 1.03 cm per pixel at 40 m, placing a person in water at 39 pixels. Fixed focus is specified deliberately: the hyperfocal distance is 4.15 m, so a lens set once is sharp from 2 m to infinity, and the aircraft never operates below 30 m. A motorised module costs more, introduces a moving part on a vibrating airframe, and can drift without indication in flight.
Radio, storage, cooling	ExpressLRS 2.4 GHz receiver, firmware 3.5 or later, with storage and mounting	8,500	In its native mode this link carries manual control and telemetry on a single autopilot port, which is why a second radio is unnecessary — a saving of approximately INR 19,000 per aircraft. The firmware version is material: the earlier mode consumes the link and would leave the aircraft with telemetry but no control.
Mesh radio	5.8 GHz node with antennas	6,000	Carries three simultaneous video streams to the ground station. Total offered load is 2.5 Mbps against a 20 Mbps channel; the constraint is link margin at range rather than bandwidth.
Motors	Tarot TL96020, 340 KV, remaining three	13,500	Procured after Phase 1 has verified the first unit.
Airframe	25 × 23 mm carbon tube, machined clamps and plate, fabricated in-house	13,000	The tube is sized such that bending stress at design thrust is approximately 4 % of material capability. Bending is not the governing failure mode; clamp design and vibration fatigue are, which is why the joints are machined rather than printed.
Battery cells	Molicel P45B 21700, 4500 mAh, 45 A, 18 units	12,600	Eighteen cells in six-series, three-parallel configuration: 292 Wh, sufficient for the mission plus a complete second search plus four minutes of loiter. Peak draw is 38.3 A per cell, so a 40 A cell would operate at 96 % of rating with no margin. Capacity and internal resistance must be matched, as a single weak cell determines pack behaviour.
Pack assembly	6S 60 A battery management system, power distribution	8,500	Cells are supplied loose. This covers the welding, fusing, balancing and monitoring required to produce

Phase 3 — Flight-test capability steps 11–12 INR 42,574

Objective: establish the ground support required to fly the first aircraft safely and repeatedly — charging, manual override and field readiness. **Released:** on airframe assembly and weigh-in. **Completion criterion:** first flight completed under safety-pilot supervision.

Item	Model	INR	Selection rationale
Battery chargers	ToolkitRC M6D, 500 W, 25 A, dual channel, 2 units	11,672	The pack is 13.5 Ah; charging within approximately one hour requires about 340 W. The commonly specified HOTA D6 Pro (INR 10,499–INR 13,500) provides 325 W per channel and would operate at its limit. The M6D provides 500 W at INR 5,836 and requires only an external supply, which the department already holds. Two units provide four channels, permitting the fleet to be charged in parallel.
Consumables	Fasteners, thread-locking compound, carbon tape, heat-shrink, printer filament	13,000	Consumed continuously during assembly and repair; interruption of supply halts build progress.
Safety-pilot radio	RadioMaster Boxer, ExpressLRS	6,955	Manual override, required for every test flight. Full-size hall-effect gimbals are specified rather than a compact unit, as assuming control of a 6.4 kg aircraft under fault conditions requires precise input.
Sun hood and monitor	Fabricated hood with second-hand monitor	1,500	Laptop displays are not legible in direct sunlight, and an operator unable to read the display cannot supervise the flight.

Phase 4 — Positioning and regulatory compliance steps 13–16 INR 66,992

Objective: establish centimetre-accurate positioning and complete statutory registration, so that geolocation accuracy can be measured on the first aircraft. **Released:** on first flight. **Completion criterion:** RTK fix demonstrated in the field and reported survivor position measured against surveyed ground truth.

Item	Model	INR	Selection rationale
Field consumables	Cables, mounts, field spares, transport protection	25,000	Position corrections are carried over the existing telemetry link, so no separate base-to-aircraft radio is required.
Base station receiver	Second GNSS receiver, same class as the airborne unit	18,000	Centimetre positioning operates by comparison against a stationary receiver on a known point. Without it the aircraft revert to standard GNSS and reported positions degrade from 0.75 m to approximately 3.9 m.
Registration	Statutory registration, three aircraft	5,000	At 6.36 kg these fall within the Small category of the Drone Rules 2021, making registration mandatory prior to flight.
Survey tripod	Photographic tripod with adapter	4,000	The base receiver requires a stable and repeatable mounting rather than survey-grade absolute accuracy, as the corrections are relative.

Phases 5 and 6 — Fleet completion steps 17–32 INR 2,02,413 each

Phase 5 objective: replicate the validated design as the second aircraft and demonstrate two-aircraft coordinated operation with verified separation. **Released:** on aircraft one completing a full mission.

Phase 6 objective: complete the fleet and demonstrate three-aircraft coordinated search under a single operator. **Released:** on two-aircraft separation being verified in flight.

Each phase repeats the Phase 2 component set exactly, in the same eight steps, at INR 2,02,413 — marginally higher than aircraft one, as all four motors are procured together. No individual step exceeds INR 29,440.

These are the final two requests by design. At the point either is made, a complete aircraft has already flown the mission, so the expenditure funds replication of a demonstrated configuration.

Summary

Phase	Objective	INR	Released on
1	Propulsion verification	31,371	Approval
2	First aircraft	1,97,238	Measured thrust
3	Flight-test capability	42,574	Airframe weighed
4	Positioning and registration	66,992	First flight
5	Second aircraft	2,02,413	Full mission flown
6	Third aircraft	2,02,413	Separation verified
	Total	**7,43,004**	

Individual steps are rounded to the rupee, so the column sums to within a few rupees of the total.

Structural cost. This arrangement carries an administrative overhead of 32 approvals, and procuring one aircraft at a time requires the same imported order to be placed three times, at a schedule cost of approximately eight to twelve weeks. Consolidating the per-aircraft phases into a single order is available if schedule becomes the binding constraint.

The detection evaluation requires no additional procurement: approximately one week on a departmental GPU and 30 GB of storage. The dataset is public and the software is open source.

7. Future work

Near term. Replace each calculated quantity with a measured one: detection accuracy, motor thrust and setup time. Evaluate near-infrared sensing, in which water absorbs strongly and human subjects do not, so a person appears bright against dark water; a camera without its infrared filter costs approximately INR 3,000 and a public dataset exists for evaluation prior to purchase. Beyond flooding, the system applies to earthquake, wildfire and avalanche search.

Longer term: autonomous aircraft under severe power constraint. The central constraint of this work — a capable aircraft deciding for itself within a very small power budget — is a hard requirement in planetary aviation. NASA's *Ingenuity* completed 72 flights on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density. *Dragonfly* launches in 2028 for Titan, where atmospheric density is four times that of Earth at one seventh the gravity.

The difficulty in those missions is not flight but that a radio command requires minutes to arrive: no operator is in the loop, so every decision is made on board. What transfers from this work is the portion that is not flood-specific — onboard survey planning, perception executed to an energy budget rather than an accuracy target, and safe behaviour in the absence of a link. One caution applies: microcontroller-class processors, frequently proposed for such missions, cannot resolve targets of the size considered here, so a planetary configuration would require a different operating point. This is a research question in its own right.

8. Conclusion

The design is complete and explicit as to which quantities are measured and which are calculated. We request the means to complete it in six phases and 32 steps, no step exceeding INR 30,000, so that no expenditure is committed before the preceding step has produced evidence.

Phase 1 is INR 31,371 and resolves the question on which the remainder depends: whether the selected motors lift this aircraft. The immediate technical requirement is smaller still — approximately one week of GPU time to evaluate the detection approach against public data.

We would welcome your guidance on two decisions: the operating altitude, which the detection evaluation will inform, and whether to adopt the two-stage detector once it has been measured.